

Nathan Yao

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Statistics and Machine Learning, Minor in Computer Science

Aug. 2023 – May 2027

- Dean's List with High Honors
- Relevant Coursework: Intermediate Deep Learning (Grad), Telecommunications Networks (Grad), Parallel and Sequential Data Structures & Algorithms, Computer Systems, Matrices & Linear Transformations

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

AppLovin

Palo Alto, CA

- Launched a playable-ad interaction feature globally that increased installs by x% and margin by x%, representing an estimated \$xxx million annualized profit increase based on backtest results.
- Built an automated system to detect and disable ad creatives that crash apps, reducing global platform post-click crash rate by x%
- Developed a Claude-assisted experimentation workflow spanning 3 coupled repositories, automating design specifications, implementation guidance, cross-repository code review, testing, and experiment analysis.

Software Engineer Intern

May 2025 – August 2025

Broadcom Inc.

San Jose, CA

- Automated testing workflows by developing and deploying a Flask Web Server with multiple endpoints that scrape experiment data for a hardware component in Apple mobile devices.
- Developed a barcode logging system and GUI to capture item-location data into MongoDB while interacting with Google Cloud APIs, enabling efficient traceability and analysis across manufacturing steps

Data Analysis Research Assistant

January 2025 – May 2025

Professor Minkyung Kim, Carnegie Mellon University

Pittsburgh, PA

- Conducted quantitative research and applied causal inference methods to analyze marketing strategies and public policy in the real estate sector
- Cleaned, merged, and standardized large-scale real estate Redfin datasets (12 million observations) from multiple structured batches for temporal and geographic data analysis

Undergraduate Teaching Assistant

January 2026 – May 2026

Carnegie Mellon University

Pittsburgh, PA

- Courses: 10-707: Advanced Deep Learning (Grad), 36-401: Modern Regression
- Taught recitations and held office hours, breaking down complex concepts into clear, approachable explanations for students

NCAA ATHLETICS: VARSITY MEN'S SOCCER

Awards: Earned 2024 & 2025 UAA All-Academic Recognition and CMU Men's Soccer Most Improved Player of 2024

Leadership Group: Facilitated open cross-functional communication among teammates and coaches, fostering a collaborative environment and aligning diverse perspectives

Soft skills: Developed strong time management, pragmatism, organizational proficiency, and a consistent focus on impact-oriented results

TECHNICAL SKILLS

Languages: Python, C, SQL, C++, Standard ML, JavaScript, HTML, R, Java

Tools and Libraries: Agentic tooling, PyTorch, NumPy, Pandas, Flask, Git, MongoDB, PostgreSQL, AWS

ADDITIONAL INFORMATION

Nationality: US Citizen; **Languages:** English (Native) and Mandarin (Bilingual)